

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JNANA SANGAMA", MACHHE, BELAGAVI-590018



Phase – 1 Project

Report on

Blockchain-based Reliable and Intelligent Donation Governance(BRIDGE)

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by

HEMANTH K S

1CD23IS058

G BHARATH

1CD23IS046

GAHAN KRISHNA K

1CD23IS049

Under the Guidance

of

Dr. Vathsala M K

Professor

Department of ISE



Department of Information Science and Engineering

CAMBRIDGE INSTITUTE OF TECHNOLOGY,

BANGALORE-560 036

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CAMBRIDGE INSTITUTE OF TECHNOLOGY

K.R. Puram, Bangalore-560 036

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING



CERTIFICATE

Certified that **Mr Hemanth K S, G Bharath, Gahan Krishna K** bearing USN **1CD23IS058, 1CD23IS046, 1CD23IS049** respectively, are bonafide students of **Cambridge Institute of Technology**, has successfully completed the Project Work Phase-I entitled "**Blockchain-based Reliable and Intelligent Donation Governance(BRIDGE)**" in partial fulfillment of the requirements for VI semester **Bachelor of Engineering in Information Science and Engineering** of **Visvesvaraya Technological University, Belagavi** during academic year 2026-2027. It is certified that all Corrections/Suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The Project report has been approved as it satisfies the academic requirements prescribed for the Bachelor of Engineering degree.

Project Guide
Dr. Vathsala M K
Professor
Dept. of ISE, CITech

Project Coordinator
Dr. Priyanka Desai,
Prof.ShivakumarM.
Dept. of ISE, CITech

Head of Department
Dr. Preethi S,
Professor & HOD,
Dept. of ISE, CITech

DECLARATION

We, **Mr Hemanth K S, G Bharath, Gahan Krishna K** bearing USN **1CD23IS058, 1CD23IS046, 1CD23IS049** of VI semester BE Information Science and Engineering, Cambridge Institute of Technology, hereby declare that the project entitled "**Blockchain-based Reliable and Intelligent Donation Governance(BRIDGE)**" has been carried out by us and submitted inpartial fulfillment of the program requirements of **Bachelor of Engineering in Information Science and Engineering** as prescribed by **Visvesvaraya Technological University, Belagavi**, during the academic year 2026- 2027.

We also declare that, to the best of our knowledge and belief, the work reported. Here does not form part of any other report on the basis of which a degree or award was Conferred on an earlier Occasion on this by any other student.

Date:

Name	USN	Signature
Hemanth K S	1CD23IS058	
G Bharath	1CD23IS046	
Gahan Krishna K	1CD23IS049	

ABSTRACT

In today's digital world, online donation platforms have become an essential medium for supporting social welfare activities, disaster relief funds, educational programs, medical emergencies, and charitable organizations. However, traditional donation systems still face several major challenges such as lack of transparency, misuse of donated funds, absence of real-time tracking, poor accountability, centralized control, and security vulnerabilities. Donors often remain uncertain whether their contributions are reaching the intended beneficiaries properly. These issues reduce donor trust and negatively impact the effectiveness of charity management systems. The proposed project, BRIDGE (Blockchain-based Reliable and Intelligent Donation Governance Engine), aims to address these problems by introducing a blockchain-based donation governance platform that ensures transparency, security, traceability, and reliability in donation management. Blockchain technology provides decentralized and immutable transaction recording, where every donation transaction is securely stored in a distributed ledger that cannot be modified or tampered with.

The BRIDGE system enables secure interaction between donors, beneficiaries, and administrators through authenticated digital transactions. The system records each donation transaction into blockchain blocks after verification, ensuring transparency and accountability throughout the donation lifecycle. Smart governance mechanisms are incorporated to monitor donation flow, prevent fraudulent activities, and improve public trust in digital charity platforms. The proposed system also supports secure authentication, transaction validation, transparent fund tracking, and immutable record management. By integrating blockchain technology with intelligent donation governance mechanisms, the BRIDGE platform provides a scalable, secure, and trustworthy solution for modern digital donation systems. This project contributes toward creating a transparent and corruption-free donation ecosystem where donors can confidently contribute funds while beneficiaries receive efficient and accountable support.

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

In recent years, digital donation platforms have become increasingly popular for supporting charitable organizations, medical emergencies, disaster relief programs, educational institutions, and social welfare activities. However, many traditional donation systems still suffer from major challenges such as lack of transparency, fund misuse, delayed transactions, poor tracking mechanisms, and limited donor trust. Donors often do not know whether their donated amount reaches the intended beneficiary completely and securely. The proposed project, BRIDGE (Blockchain-based Reliable and Intelligent Donation Governance Engine), is designed to solve these issues using blockchain technology and intelligent governance mechanisms. The system provides a transparent, secure, decentralized, and tamper-proof donation management platform. Every donation transaction is recorded in blockchain blocks, making the information immutable and traceable.

The BRIDGE system enables donors, administrators, and beneficiaries to interact through a secure platform where each transaction is validated and stored permanently. The project improves accountability by ensuring that donation records cannot be altered or deleted. Smart governance mechanisms also help in monitoring fund utilization and preventing fraudulent activities. The proposed system aims to increase public confidence in online donation platforms by ensuring transparency, security, and reliability throughout the donation process..

1.2 RELEVANCE OF PROJECT

The rapid growth of online financial transactions and crowdfunding platforms has increased the demand for secure and transparent donation systems. Traditional charity systems usually rely on centralized databases and manual verification methods, which are vulnerable to corruption, hacking, and manipulation.

The BRIDGE project is highly relevant in the modern digital era because:

- Blockchain technology provides transparency and immutability. Donors can verify where and how their funds are utilized. Fraudulent transactions and unauthorized

modifications can be reduced. The system improves trust between donors and charitable organizations. Real-time transaction tracking ensures accountability.

- Secure digital records eliminate dependency on paper-based management. The project supports digital transformation in social welfare sectors. The project is especially useful for NGOs, hospitals, disaster management organizations, crowdfunding agencies, and social service institutions that require trustworthy donation management.

1.3 PROBLEM STATEMENT

Existing online donation systems face major issues related to transparency, security, and trust. Donors often cannot track how their funds are used, leading to reduced confidence. Most platforms use centralized databases, which are vulnerable to cyberattacks, data manipulation, and single-point failures.

There is also a risk of corruption and unauthorized modification of donation records, as these systems lack a tamper-proof transaction history. Additionally, tracking donations and verifying transactions is often slow and inefficient, causing delays and reducing accountability. Therefore, there is a need for a secure, transparent, and decentralized donation system that ensures reliable transaction tracking, prevents data manipulation, and improves donor trust.

1.4 OBJECTIVES

The main objectives of the our project are:

- To develop a secure and decentralized blockchain-based donation system that ensures safe transactions and strong authentication mechanisms.
- To provide transparency and accountability by enabling donors to track transactions and ensuring clear interaction between donors, administrators, and beneficiaries.
- To maintain immutable and tamper-proof records of all donation activities, reducing fraud and unauthorized modifications.
- To design a scalable and efficient system that supports real-time tracking, effective fund governance, and future enhancements.

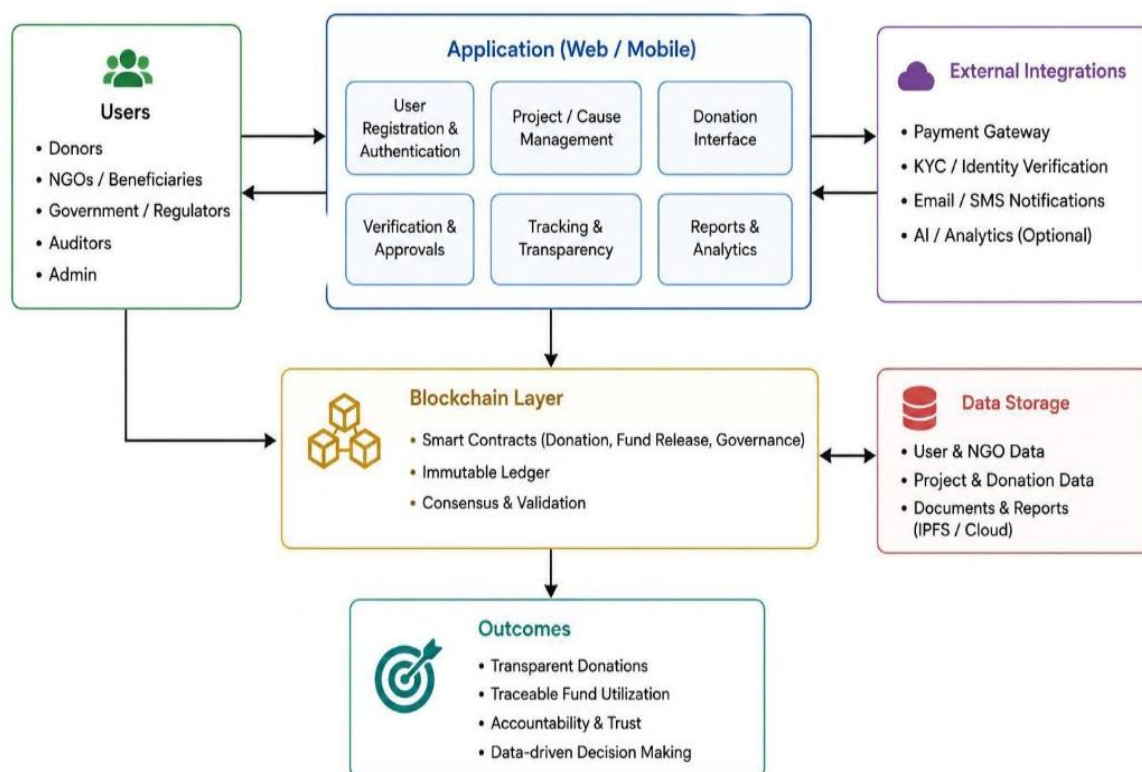
1.5 METHODOLOGY

The BRIDGE project follows a systematic software development methodology to ensure proper planning, implementation, testing, and deployment.

Step 1: Requirement Analysis

In this phase, the existing donation systems are analyzed to identify their limitations related to transparency, security, and accountability. Functional and non-functional requirements are gathered for the proposed blockchain-based donation platform.

Step 2: System Design



Step 3: Blockchain Integration

Blockchain technology is integrated to record donation transactions securely. Each donation transaction is verified and stored as immutable blocks connected using cryptographic techniques.

Step 4: Application Development

The frontend and backend modules are developed using appropriate technologies. User interfaces for donors, administrators, and beneficiaries are implemented.

Step 5: Database Management

Database systems are configured to manage user data, transaction metadata, authentication details, and system logs.

Step 6: Testing and Validation

The system is tested for:

- Functional correctness
- Security vulnerabilities
- Transaction integrity
- Blockchain validation
- User authentication
- Performance and reliability

Step 7: Deployment and Maintenance

The final system is deployed and monitored regularly. Security updates, bug fixes, and future enhancements are performed during maintenance.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

A literature survey is conducted to analyze existing donation management systems, blockchain-based financial platforms, and security mechanisms in digital transactions. It highlights key limitations such as lack of transparency, security vulnerabilities, and inefficient tracking in current systems. Blockchain technology, with features like decentralization, immutability, and secure record management, has been widely proposed to address these issues. The survey primarily focuses on online donation platforms, smart contracts, secure transaction handling, fraud prevention, and decentralized governance models to identify effective solutions and improvements.

2.2 RELATED WORKS

[1] Singh, R., & Patel, A. (2023). Blockchain-based System for End-to-End Donations Monitoring. In *International Conference on Advanced Computing and Innovative Technologies (ACIT)* (pp. 1–6). IEEE. <https://doi.org/10.1109/ACIT58888.2023.10453793>

This paper presents a blockchain-based donation monitoring system designed to improve transparency and accountability in charitable transactions. The proposed system records every donation transaction in an immutable blockchain ledger, allowing donors to track the complete flow of funds from donation to beneficiary distribution. Smart contracts are used to automate transaction validation and reduce manual intervention. The study highlights how blockchain technology minimizes corruption, prevents unauthorized data modification, and improves donor trust through transparent and secure donation management.

[2] Sharma, P., & Verma, N. (2024). Blockchain-Powered Charity Integrity System. In *International Conference on Emerging Technologies in Computer Science (ICETCS)* (pp. 1–7). IEEE. <https://doi.org/10.1109/ICETCS61022.2024.10543760>

This paper introduces a blockchain-powered charity management framework aimed at ensuring integrity and transparency in online donation platforms. The system uses decentralized ledger technology to maintain immutable donation records and provide real-time visibility into fund utilization. Smart contracts automate approval and verification processes, reducing fraud and administrative overhead. Experimental analysis shows that blockchain-based governance improves accountability, donor confidence, and operational efficiency in charitable organizations.

[3] Alharbi, M., Khan, S., & Joseph, T. (2024). Smart Blockchain Networks: Revolutionizing Donation Tracking in Web 3.0. Computer Networks, 245, 110381. <https://doi.org/10.1016/j.comnet.2024.110381>

This paper explores the application of Web 3.0 and blockchain technologies in modern donation tracking systems. The proposed framework leverages decentralized smart blockchain networks to provide secure, transparent, and traceable donation transactions. The study discusses how smart contracts and distributed ledgers eliminate dependency on centralized authorities while improving trust among stakeholders. The paper also highlights the role of blockchain in enabling real-time monitoring and governance of charitable funds within decentralized ecosystems.

[4] Chen, Y., & Li, X. (2024). Eisdspa: An Efficient and Secure Blockchain-Based Donation Scheme With Privacy Protection and Auditability. IEEE Open Journal of the Communications Society, 5, 1–14. <https://doi.org/10.1109/OJCOMS.2024.3504403>

This paper proposes a secure blockchain-based donation scheme that focuses on privacy protection, transaction auditability, and efficient donation management. The framework combines cryptographic techniques with blockchain architecture to ensure secure fund transfer and prevent unauthorized access to sensitive donor information. The system enables transparent auditing while preserving donor privacy through encryption mechanisms. Experimental evaluations demonstrate improved security, reliability, and scalability compared to traditional centralized donation systems.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING TECHNOLOGY

Existing online donation systems mainly use centralized architectures where all data is stored and managed by a central authority. These systems provide basic donation functionalities such as user registration, payment processing, and fund transfer. However, traditional systems face multiple security and transparency challenges.

3.1.1 Features of Existing Systems

- Online payment integration
- User registration and authentication
- Donation request management
- Centralized database storage
- Basic transaction history
- Administrative control mechanisms

3.1.2 Limitations of Existing Systems

- Lack of transparency in fund utilization
- Risk of corruption and fraud
- Centralized database vulnerabilities
- Poor accountability mechanisms
- No immutable transaction history
- Limited donor trust
- Difficulty in transaction auditing
- Risk of unauthorized data modification
- Possibility of fake donation campaigns
- These problems highlight the need for decentralized and blockchain-based solutions.

3.2 PROPOSED SYSTEM

Our project introduces blockchain technology into donation management to provide a transparent, secure, and decentralized donation governance platform. The system records every donation transaction in blockchain blocks after validation. Each block is linked cryptographically, ensuring transaction immutability and preventing unauthorized modifications.

3.2.1 Key Features

- Blockchain-based transaction recording
- Transparent donation tracking
- Immutable ledger maintenance
- Secure authentication system
- Real-time transaction monitoring
- Decentralized governance mechanism
- Fraud prevention through validation
- Beneficiary verification system
- Smart transaction management
- Donation history tracking

3.2.2 Advantages of Proposed System

- High transaction security
- Increased donor trust
- Tamper-proof data storage
- Better accountability
- Improved transparency
- Reduced corruption risks
- Faster verification processes
- Reliable digital governance
- Secure access control
- Efficient auditing capabilities

3.2.3 Working Procedure

- Donor registers into the system.
- User authentication is performed.

- Donation transaction is initiated.
- Transaction is validated using blockchain mechanisms.
- Validated transaction is stored in blockchain blocks.
- Administrators monitor and approve fund allocation.
- Beneficiaries receive verified funds.
- Donors can track transaction status transparently.

3.3 HARDWARE REQUIREMENTS

The BRIDGE project mainly focuses on software and blockchain technologies therefore, minimal hardware resources are required.

3.3.1 Hardware Components

Processor

A multi-core processor is required to handle blockchain simulation, backend processing, database operations, and frontend rendering efficiently.

RAM

At least 4 GB RAM is required for running development environments, blockchain platforms, and web applications simultaneously.

Storage

Sufficient storage is required for project files, blockchain ledgers, databases, and application modules.

Internet Connectivity

Stable internet connectivity is necessary for blockchain synchronization, cloud services, and web-based application testing.

3.4 SOFTWARE REQUIREMENTS

The software requirements for the BRIDGE project include programming tools, blockchain platforms, databases, and development environments.

3.4.1 Software Components

Ethereum Blockchain

Ethereum is used for creating decentralized blockchain networks and executing smart contracts securely.

Ganache

Ganache provides a local blockchain environment for testing and development.

Solidity

Solidity is used to develop smart contracts that automate transaction validation and governance functions.

React.js

React.js is used to build responsive and interactive frontend interfaces.

Node.js

Node.js handles backend services, APIs, transaction processing, and blockchain communication.

MongoDB

MongoDB stores user information, transaction metadata, and application records.

CHAPTER 4

MODELING

4.1 PURPOSE

The primary purpose of the BRIDGE project is to create a transparent, secure, and decentralized donation governance platform using blockchain technology. The system aims to eliminate corruption, improve accountability, and increase donor trust by maintaining immutable donation records.

Specific Goals of the Project

- Real-time donation tracking
- Secure blockchain-based transactions
- Transparent fund utilization
- Prevention of fraudulent activities
- Efficient governance mechanisms
- Secure user authentication
- Beneficiary verification
- Reliable auditing capabilities
- Digital transformation of charity systems

4.1.1 Importance of the Project

With increasing online financial activities, ensuring trust in donation platforms has become essential. The BRIDGE project provides a reliable framework for secure digital charity management while improving public confidence in online donations.

4.2 LEVELS OF SOFTWARE DESIGN

1. Architectural Design

The architectural design defines the complete structure of the BRIDGE system.

Main Modules

- User Module
- Donor Module
- Beneficiary Module
- Admin Module
- Blockchain Module
- Authentication Module
- Transaction Verification Module
- Reporting Module

The architecture follows a modular design where each module performs specific responsibilities independently.

2. High-Level Design (HLD)

The High-Level Design describes communication between modules.

- Users interact through web interfaces.
- Authentication verifies user identity.
- Donation requests are processed through blockchain validation.
- Transaction records are stored in immutable blocks.
- Administrators monitor and verify fund distribution.
- Database stores metadata and system logs.

3. Low-Level Design (LLD)

The Low-Level Design describes detailed implementation.

Functional Components

- User registration logic
- Login authentication
- Blockchain transaction generation
- Smart contract execution and Donation verification
- Transaction history generation
- Report generation
- Database operations

- Database operations
- API request handling

Each module is implemented independently to ensure scalability and maintainability.

4.3 SCOPE

The system is designed with the following functional and non-functional scope:

4.3.1 Functional Scope

- The BRIDGE project supports:
- Online donation processing
- Blockchain transaction storage
- Real-time donation tracking
- Donor and beneficiary interaction
- Secure authentication
- Fund verification mechanisms
- Digital governance management
- Transaction auditing

4.3.2 Non-Functional Scope

- High security
- Scalability
- Reliability
- Transparency
- Performance efficiency
- User-friendly interfaces

4.3.3 Future Scope

The future enhancements include:

- AI-based fraud detection
- Mobile application support
- Multi-currency donation systems
- Cloud deployment
- Government welfare integration
- Smart analytics dashboard
- Advanced blockchain consensus mechanisms
- Integration with digital identity systems

4.4 SYSTEM ARCHITECTURE

The BRIDGE system architecture consists of multiple layers that work together to provide secure and transparent donation management.

Presentation Layer

Handles user interaction through web interfaces for donors, administrators, and beneficiaries.

Application Layer

Processes business logic including donation management, authentication, and transaction handling.

Blockchain Layer

Maintains immutable transaction records and validates donation transactions.

Database Layer

Stores user details, transaction metadata, logs, and application information.

Security Layer

Provides authentication, encryption, authorization, and access control.

Workflow of the System

1. User accesses the donation platform.
2. Authentication verifies the user.
3. Donor initiates donation transaction.
4. Blockchain validates the transaction.
5. Transaction is stored securely.
6. Administrators verify fund allocation.
7. Beneficiaries receive approved donations.
8. Donors track transaction history transparently.

The architecture ensures transparency, security, scalability, and accountability.

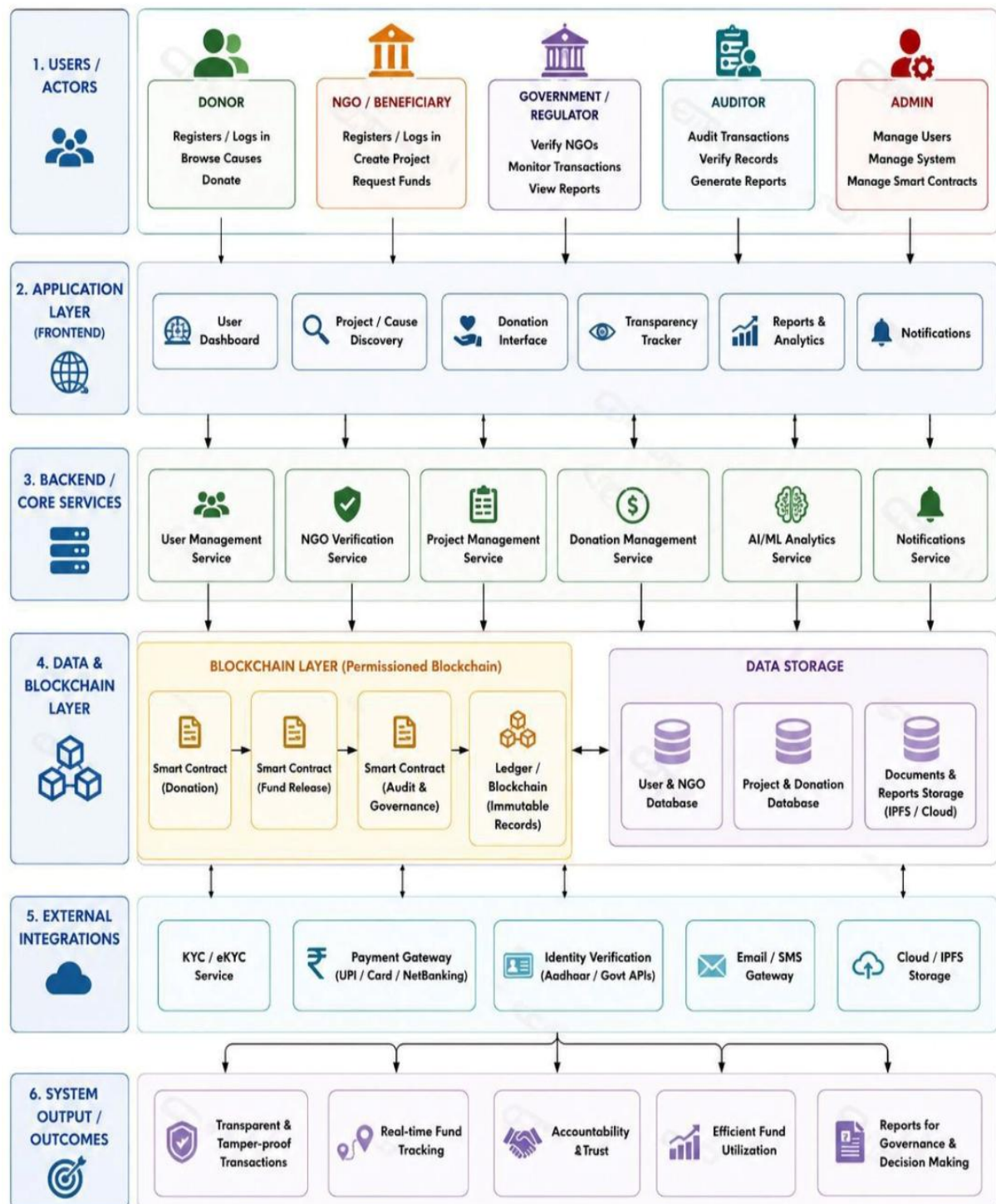


Fig 4.1 System Architecture

4.5 USE CASE DIAGRAM

4.5.1 Actors

- Donor
- Administrator
- Beneficiary
- Blockchain Network

4.5.2 Donor Use Cases

- Register/Login
- Make Donation
- Track Donation Status
- View Transaction History
- Verify Fund Utilization

4.5.3 Administrator Use Cases

- Verify Transactions
- Manage Beneficiaries
- Approve Donations
- Monitor Blockchain Records
- Generate Reports

4.5.4 Beneficiary Use Cases

- Submit Requests
- Receive Funds
- View Donation Status

4.5.5 Blockchain Use Cases

- Validate Transactions
- Store Immutable Records
- Maintain Distributed Ledger

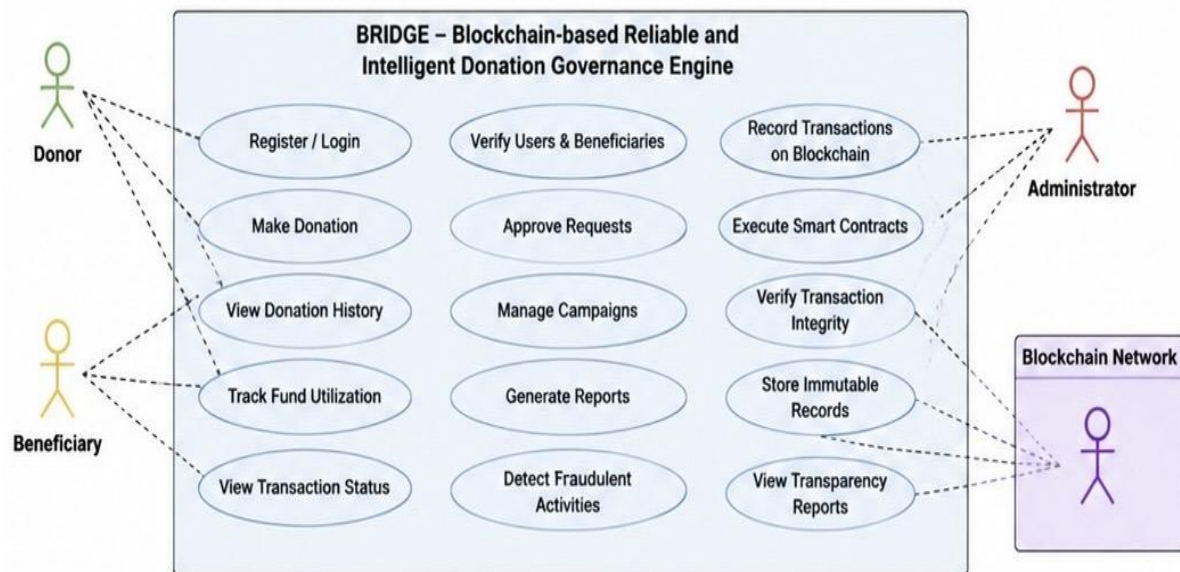


Fig 4.2 Use Case Diagram

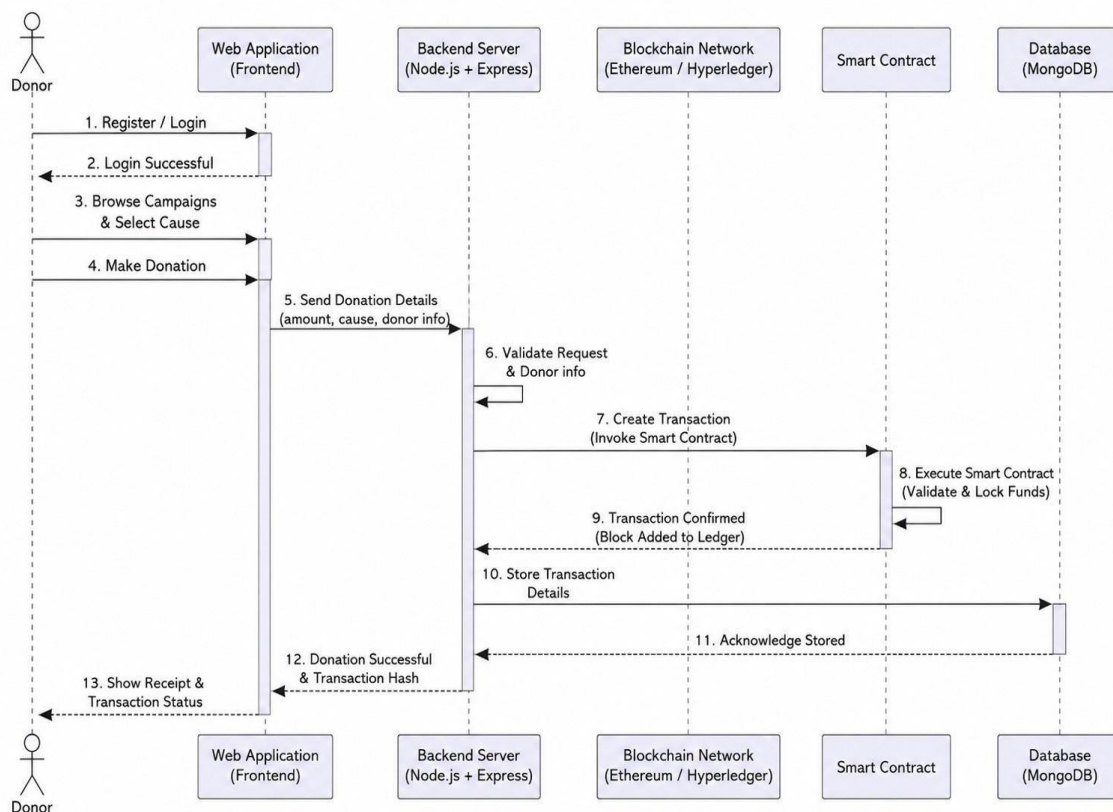
4.4 SEQUENCE DIAGRAM

The sequence diagram explains interaction among users, application modules, and blockchain components.

Sequence Flow

1. User logs into the system.
2. Authentication module validates credentials.
3. Donor initiates donation transaction.
4. Transaction request is sent to blockchain.
5. Blockchain validates transaction integrity.
6. Valid transaction is added to blockchain.
7. Database stores transaction metadata.
8. Administrator monitors transactions.
9. Beneficiary receives approved funds.
10. Donor views transparent transaction history.

The sequence diagram demonstrates secure and transparent communication between all system components.



4.3 Sequence diagram

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- [3] T. Li, M. Hua, and X. Wu, “A hybrid CNN-LSTM model for forecasting particulate matter (PM2.5),” IEEE Access, vol. 8, pp. 26933–26940, 2020
- [4]T. H. Putri et al., “Fine-Tuning of Predictive Models CNN-LSTM and CONV-LSTM for Nowcasting PM2.5 Level,” IEEE Access, vol. 12, pp. 28988–28999, 2024
- [5] L. S. and A. Krishnamoorthy, “Deep Learning Approaches for Air Quality Forecasting: An LSTM and CNN Perspective,” in Proc. 2nd Int. Conf. IoT, Commun. Automat. Technol. (ICICAT), 2024.